

4.6 Detailed specific example: Tonga



Tonga is a small archipelago located in Oceania, in the South Pacific Ocean, approximately two-thirds of the way between New Zealand and Hawaii. The islands can be divided into two groups. There are the Eastern islands which are formed from a limestone base, and the Western Islands that have volcanic origins. The volcanic islands lie along a subducting plate boundary, with the Pacific plate subducting beneath the Indo-Australian plate.

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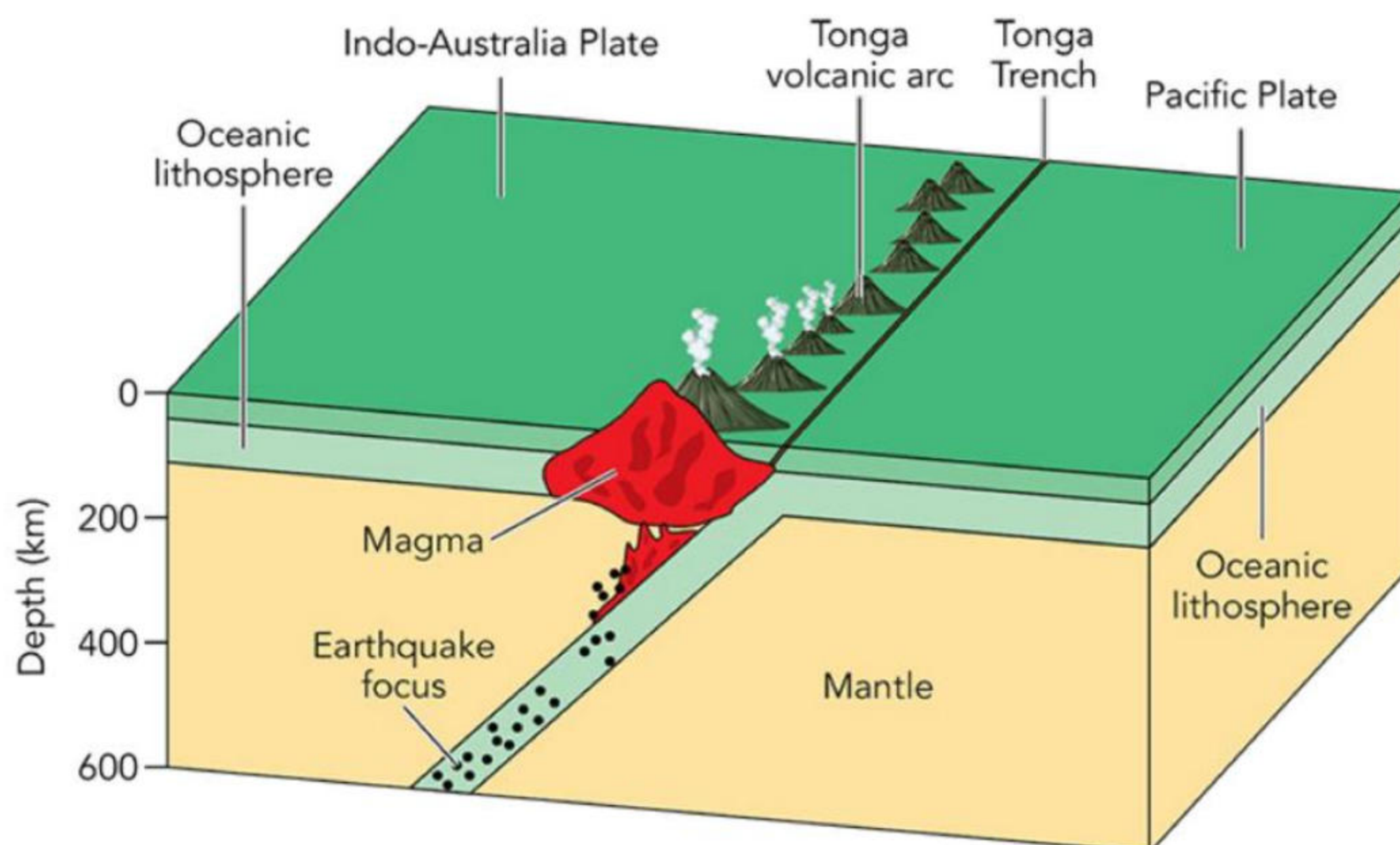
In December 2021, a submarine volcano (a volcano under the ocean), in the Tongan islands, named Hunga Tonga-Hunga Ha'apai, started to erupt. In the early stages of the eruption, the island increased in size by 300–600 metres on its eastern side. Before the main eruption, which occurred four weeks later, ongoing and increased earthquake activity and steam emissions were recorded for the volcano.

On 15 January 2022, the main eruption occurred. It was very large and rated as a 5.7 on the Volcanic Explosivity Index. The eruption generated the largest atmospheric explosion ever recorded by modern instruments: 10 km³ of material was displaced, and approximately 140 megatons of water vapour was ejected into the atmosphere.

Causes

The Tonga island chain is formed along a subducting plate boundary (Source A). During the process of subduction, the sinking plate melts due to pressure, friction, and heat. This results in the formation of magma which is forced to the surface to form volcanic islands. With the Hunga Tonga-Hunga Ha'apai eruption, the magma inside the volcano was under enormous pressure. The shifting Earth pulled the surrounding sea water into the subduction zone. As the seawater met the hot magma, it boiled, adding water vapour and other gases to the magma chamber. When the rock fractured under the pressure, the gas expanded, blasting the magma apart in a massive eruption that sent a plume of material 58 km into the atmosphere.

With the crater located about 200 metres below sea-level, water poured into the volcano, turning the water into steam. This added to the rapid expansion of the eruption.



Source A: The subduction occurring between the Pacific plate and the Indo-Australian plate has resulted in an island arc that forms the Tongan archipelago

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- a Name the type of plate boundary between the Pacific and Indo-Australian Plate.
- b How deep under the crust is the magma forming?
- c Explain how a chain of volcanoes has formed along the Tonga trench. Consider the type of plate boundary that this type of volcano is found on.

The impacts of the eruption

The 2022 eruption was so large that it had a number of impacts, both locally and internationally. The scale of the eruption surprised scientists, despite the predictions that the volcano was going to erupt.

Volcanic impacts

The eruption created the following impacts:

- Tsunamis around the Pacific basin, affecting distant areas such as Far Eastern Russia, Chile, and Peru.
- The tsunami was up to 15 metres high in Tonga.
- A large plume of ash, 5 km wide and 58 km high, was sent into the atmosphere. This drifted north and north-east and settled on surrounding islands.
- Steam plumes, ash, and steam bursts were recorded. The main plume reached up to 58 km into the atmosphere.
- The explosion was heard 2000 km away in New Zealand, two hours after the eruption.
- The explosion caused an unexpected hazard of atmospheric shockwaves that moved around the planet. These drove tidal surges causing damage on distant coastlines. The shockwaves were so large that they were visible on satellite images.
- Earthquakes and tremors shook buildings in Tonga.

Social and economic impacts

- Four people on the islands died and several people were injured.
- For a short period after the eruption, people struggled to breathe due to the ash in the air.
- Flooding and damage to properties on the islands from the Tonga tsunami, with some towns and resorts severely damaged.
- The total economic cost caused by the eruption was estimated to be \$182 million, more than 36% of the Tonga GDP.
- Air traffic was interrupted as it became unsafe for planes to fly. First-aid planes were only able to access the island five days after the event, once the ash had cleared sufficiently to allow planes to fly.
- Communication channels were damaged, including the main undersea cable (necessary for internet connectivity). Phone lines were only partially restored five days after the event.

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Environmental impacts

Due to the scale of major volcanic eruptions, they can affect the atmosphere as well as local physical geography and the surrounding oceans. Large volcanic eruptions can eject sulphur dioxide into the atmosphere, which reflects sunlight and can cause a temporary cooling effect. However, during the Hunga Tonga-Hunga Ha'apai eruption, the volcano produced large amounts of water vapour which was ejected into the atmosphere. Water vapour is a greenhouse gas and traps heat in the atmosphere, negating the cooling effect.

The ejected ash affected the surrounding air and sea in several negative ways. The ash contributed to acid rain and acid gas, affecting surface water and groundwater. There is concern in the scientific community that this acidification could cause long-lasting damage to coral reefs, eroding coastlines, and disrupting the life cycles of aquatic organisms. Falling ash can also coat coral reefs, affecting an ecosystem which is already threatened by human activity. The eruption also added iron (originating from the magma) to the water. Increased iron levels affect the food web, triggering the growth of blue-green algae and sponges, which further degrade the reefs.

Positive impacts included the addition of nutrient-rich ash to the soils. In the long term, this will increase soil fertility and support plant growth. This increases habitat availability for terrestrial organisms.



Source **B**: Hunga Tonga-Hunga Ha'apai volcano on the 6 January 2022 before the eruption



Source **C**: Hunga Tonga-Hunga Ha'apai volcano on the 17 January 2022 just before the eruption

TASK 2: Study Sources **B** **C**

a What type of volcano is Hunga Tong-Hunga Ha'apai?

b Sources B and C show satellite images of the Hunga Tonga-Hunga Ha'apai volcano, before and just before the eruption in 2022. In pairs, **discuss what changes you observe between Sources B and C**. Describe the changes that you observe in a brief summary of 50 to 100 words.

c **Draw a sketch diagram** showing the changes from before and after the eruption. Annotate the diagram to explain the changes.

d How useful are satellites in monitoring volcanic activity?

TIP

When drawing a sketch diagram, keep the lines clear and neat. Ensure that you label the most important features. When completing a comparative sketch, use annotations to explain change.

Responses to the eruption

Several different steps were taken to protect people from the impacts of the eruption. Table 4.5 gives some of the key actions taken.

Table 4.5: The emergency response to the eruption of Hunga Tonga-Hunga Ha'apai

	Date	Action taken
Early warnings	December 2021	The Tongan government notified the local population of increased volcanic activity.
	14 January	A tsunami warning was issued to ensure that people moved away from the coastline on surrounding islands.

Immediate responses	2022	
	15 January 2022	Another tsunami warning was issued after the main eruption. People were warned to stay away from shorelines and low-lying areas were evacuated.
	15–17 January 2022	People were urged to stay indoors to avoid the ash which can irritate eyes and increase the risk of medical conditions like asthma and bronchitis. People were advised to prevent contamination of freshwater storage with sulphur dioxide by covering rainwater tanks. Drinking water became more acidic as time passed.
	20 January 2022	First aid planes from New Zealand and Australia arrived and phone lines were partially restored.
	2 February 2022	After receiving aid shipments, the country went into COVID-19 lockdown when two port workers tested positive.
International assistance	• New Zealand, China, the USA, and other countries provided relief supplies to Tonga to assist with the reconstruction. • New Zealand provided desalination equipment to produce fresh water. • Australia carried out surveillance flights to assess damage. • France, the USA, Australia, and New Zealand co-ordinated humanitarian aid. • Engineers worked to restore internet access to re-establish communications.	

TASK 3

- Review the steps that were taken to protect the population. Explain how this helped to keep the death toll low during this volcanic event.
- Desalination is the removal of salt from salt water to create fresh water. Explain why the provision of desalination equipment by New Zealand was so important to the island population.
- Explain how restoration of communication plays an important role in the emergency response to a natural disaster like the Hunga Tonga-Hunga Ha’apai eruption.

Managing the impacts of eruptions in Tonga

There are a number of strategies in place in Tonga to help manage the impacts of eruptions. These strategies were employed in the 2021 eruption, but many of the techniques sitting under each of these strategies have since been improved.

Prediction and monitoring

The island chains are constantly under observation for change in volcanic and seismic activity, for example, by using seismographs and satellite imagery. There is an early warning system fed by information from these technologies, which gives the population time to react when a volcano starts to show a change in activity. Hazard mapping is also used, mapping areas at highest risk of tsunami, earthquakes, and ash fall.

In late 2023, Tonga secured funding through the Climate Risk and Early Warning Systems initiative (run in part by the World Meteorological Organization and the World Bank) to strengthen its early warning system and ensure faster and broader dissemination of warnings across the island chain.

Planning and protection

Predicting tectonic activity alone will not protect the population against hazards. The following techniques are used in Tonga to plan for eruptions and protect people from their effects:

- Developed evacuation routes and plans for low-lying areas and islands. Tsunami evacuation maps can be seen around the main island to raise public awareness and educate people on the safest route inland.
- Well-developed emergency plans and briefing of the community. For example, in November 2023, over 1500 school students participated in a drill to test the early warning systems.
- Stores of items such as food, water, shelter, clothing, and medical resources for response in the event of an eruption.

The eruption of Hunga Tonga-Hunga Ha’apai, and the resulting tsunami, caused significant damage to housing and infrastructure on Tonga. Once the first stage of clearing up is completed, the focus has to turn to rebuilding. In some instances, it has to turn to relocating families that have lost everything.

The rebuild must take place in a way that makes houses stronger, safer, and more resistant to damage in the event of another natural disaster. However, the urgent need for housing, and the difficulty of securing building materials due to pandemic-related supply chain issues, has limited the ability of the islands to

build safe, robust homes.

Source **D**: News article on Tonga's recovery

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TASK 4: Study Source **D**

- a** Explain why the COVID-19 lockdown would have slowed down the recovery of the islands after the eruption. Consider if the dependence on outside assistance for help is sustainable.
- b** In pairs, discuss the type of features that should be included in the building of homes to make them safer during volcanic eruptions. List your three main ideas.

IMPROVE THIS ANSWER

Explain why volcanoes are likely to form along a destructive plate boundary.

Two plates move towards each other 1, as one plate sinks 2 under the other, they rub together 3 and one melts. This results in magma forming, which is pushed to the surface through cracks and forms a volcano 4. This causes a volcano to erupt, which can damage houses when the ash lands on the houses. It can also cause death or injury 5.

1 The answer should use the correct term for the plate boundary. Here it is 'converging' and a 'destructive plate boundary'.

2 Using technical terminology would make this answer stronger, for example, 'subduction' here rather than 'one plate sinks under the other'.

3 The answer needs to be more specific here: friction and heat is causing the melting of the plates.

4 More detail is required here. The answer should discuss how pressure builds up as the magma accumulates below the surface, and how this then causes the material to be forced to the surface.

5 This is beyond the scope of the question. The question asks why volcanos form along a destructive plate boundary, not what the impacts are.

Now, write an improved response to this answer using the guidance above.

SELF-EVALUATION CHECKLIST

After studying this topic, think about how confident you are with the different topics. This will help you to find any gaps in your knowledge and learn more effectively.

I can:	Needs more work	Getting there	Ready to move on
describe the characteristics of the Earth’s layers, tectonic plates, volcanoes, and earthquakes			
explain the main hazards of earthquakes and volcanoes			
give reasons for why people live in earthquake zones			
explain the causes and characteristics of earthquakes			
give the impacts of earthquakes and volcanoes on people, the economy, and the environment			
explain strategies for managing the risks from earthquakes and volcanoes.			



> Topic 5

Climate change

LEARNING INTENTIONS

By the end of this topic you will be able to:

- outline the natural and human causes of climate change
- explain the effects of climate change at a local, regional, and global scale
- assess the strategies to manage the impacts of climate change.

GETTING STARTED

In pairs, discuss the following questions:

- 1 What is your understanding of climate change? How is it different to global warming?
- 2 What do you think are the causes of climate change? Can you categorise these causes?
- 3 How might climate change affect different aspects of the environment? How might it affect people? Draw a concept map that links the causes you identified earlier with these impacts.

Share your ideas with the class.